

Dual-Functional TiN/BN@separator for enhanced electrochemical performance and safety of lithium-ion batteries

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Abstract: As the energy density of lithium-ion batteries continues to increase, the performance requirements for separators have become more stringent. Traditional polyolefin separators offer advantages such as low cost and excellent chemical stability. But they suffer from poor electrolyte wettability, thermal stability, and high interfacial resistance, which hinder the development of high-performance lithium-ion batteries. To address these issues, this study introduces a co-blending coating method using titanium nitride and boron nitride particles to enhance the overall performance of polypropylene separators. The TiN/BN@PP-15 separator demonstrates superior wettability, thermal stability, and electrochemical performance. Under a high discharge rate of 5 C, it achieves an initial discharge capacity of 130.7 mAh/g. Additionally, batteries assembled with the TiN/BN@PP-15 separator exhibit the lowest interfacial resistance of 79.7 Ω , a high ionic conductivity of 0.466 mS cm⁻¹, and a lithium-ion diffusion coefficient of 11.7×10^{-14} cm²s⁻¹. These results indicate that the co-blending coating of TiN and BN particles not only improves the physicochemical properties of PP separators but also ensures high discharge capacity and cycling stability under high-rate conditions, providing reliable technical support for high energy density and high rate energy storage systems.

Keywords: Electrolyte wettability; Thermal stability; Co-blending coating; Electrochemical performance; High discharge capacity

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1. Introduction

With the rapid development of electric vehicles, portable electronic devices, and renewable energy storage, lithium-ion batteries have become crucial energy storage devices on a global scale. Usually, they consist of a positive electrode, a negative electrode, an electrolyte, a separator, and other components[1-4]. As one of the essential components, the separator is often referred to as the "third electrode" and plays a crucial role in determining the stability and performance of the battery. The separator primarily serves to physically isolate positive and negative materials to prevent short circuits while permitting the free passage of lithium ions to maintain the electrochemical equilibrium within the battery and ensure its safety and performance[5, 6]. Although the separator remains electrochemically inert and refrains from engaging in redox processes, it exerts a critical influence over the stability of the interface between electrolytic solutions and electrode materials, and its performance significantly influences the cycle and rate capabilities of the battery[7]. Traditional polyolefin separators suffer from inadequate wettability, large interfacial impedance, and poor thermal stability, which can easily trigger short circuits and lead to serious safety concerns[8-10]. Therefore, the development of a high-performance separator that can simultaneously enhance electrochemical performance and guarantee safety has emerged as a critical challenge for researchers.

Polyolefin separators, such as polypropylene (PP) and polyethylene (PE) separators, dominate the market due to their excellent mechanical properties and chemical stability. However, they suffer from poor electrolyte wettability and low ionic conductivity, which hinder their application in high-energy-density storage systems[11-13]. To address these issues, researchers have attempted surface modifications by introducing ceramic particles such as silicon dioxide (SiO_2) and aluminum oxide (Al_2O_3) onto the surface of the separator[14-17]. Although this approach improves wettability and thermal stability to some extent, challenges remain, including the uneven distribution of ceramic particles and interface discontinuities that can negatively impact the overall performance of the separator. In addition, the inherent volume expansion and low electrical conductivity of some oxides may lead to electrochemical instability during battery operation. In contrast, nitride materials have demonstrated superior electrochemical stability in lithium-ion battery separators, which make them promising candidates for modification[18]. Current nitride modification methods focus on surface coating, where nitride particles such as aluminum nitride (AlN) and boron nitride (BN) are applied to commercial separators to enhance resistance against lithium dendrite formation[19, 20]. However, simple surface

modifications do not fundamentally improve the comprehensive performance of separators, and limitations in thermal stability and high-rate discharge performance remain[21].

To overcome these shortcomings, this study proposes a novel modification method that combines inorganic materials blending with surface coating to achieve a comprehensive enhancement of separator performance[22]. By uniformly blending inorganic materials and coating them onto the separator surface, the microstructure and physicochemical properties of the separator are optimized. Leveraging the excellent electrical conductivity of titanium nitride (TiN) and the outstanding thermal stability of boron nitride (BN), a composite separator was developed by co-coating TiN and BN particles onto the surface of traditional PP separators[23, 24]. Experimental results indicate that the modified separator exhibits significantly improved thermal stability, greatly reducing the risk of battery short circuits induced by high temperatures and enhancing overall safety. Furthermore, the electrolyte wettability is markedly enhanced, promoting uniform electrolyte infiltration and ion transport, thereby achieving substantial enhancements in the rate capability and cyclability of the battery. In contrast to batteries assembled with traditional PP separators, those using the TiN/BN@PP separator demonstrate superior electrochemical performance, achieving a 21% higher discharge capacity at 5 C. This research provides a practical strategy for designing high-performance lithium-ion battery separators[25].

2 Experimental section

2.1 Fabrication of TiN/BN@PP separator

In this study, the TiN powder and BN powder used were purchased from Aladdin Reagent Co., Ltd both with a purity of 99.9%. A Celgard 2400 PP separator (25 μm thick) was used as the basic separator for modification. During the preparation process, TiN powder, BN powder, a conductive agent (Super P), and a binder (PVDF) were added to a mortar in a mass ratio of 8.5:0.5:0.5:0.5 for an initial mixing. After 20 minutes of dry mixing, N-methyl-2-pyrrolidone (NMP) was gradually added to form a homogeneous slurry. The slurry was uniformly coated onto one side of the PP separator using a coating speed of 2 mm/s. The coated separator was dried in a vacuum oven at 60 $^{\circ}\text{C}$ for 12 hours, then cut into 19 mm diameter discs for subsequent use. Based on coating thickness, the modified separators were labeled as TiN/BN@PP-5, TiN/BN@PP-10, and TiN/BN@PP-15, while an unmodified PP separator was employed as the control sample.

2.2 Characterization and measurements of TiN/BN @PP separator

The surface morphology of the TiN/BN@PP separator was examined using a Hitachi SU8010

scanning electron microscope (SEM, Japan) at an acceleration voltage of 10 kV. Wettability characterization was conducted through three metrics: contact angle measurements, porosity tests, and electrolyte uptake rate. Porosity was determined by immersing samples in n-butanol (C₄H₉OH) for 2 hours following Equation (1)[26]:

$$Porosity(\%) = \frac{W_1 - W_0}{\rho \times V} \times 100\% \quad (1)$$

Where, W_0 and W_1 are the weights of the separator before and after soaking, ρ is the density of C₄H₉OH, and V is the separator's volume. Electrolyte uptake ratio was measured via gravimetric analysis by comparing separator masses before and after immersion in the electrolyte for 2 hours, and computed according to Equation (2)[27]:

$$Electrolyte\ uptake(\%) = \frac{M_1 - M_0}{M_0} \times 100\% \quad (2)$$

Where, M_0 and M_1 are the masses of the separator before and after soaking in the electrolyte, respectively. The thermogravimetric analysis, thermal shrinkage and flame retardant properties of the separators were assessed to determine their thermal stability. The area shrinkage rate was calculated using Equation (3)[28]:

$$Shrinkage(\%) = \frac{S_0 - S}{S_0} \times 100\% \quad (3)$$

Where, S_0 and S are the areas of the separator before and after heating.

2.3 Electrochemical measurements

This study fabricated CR2025 cells for comprehensive electrochemical characterization. All cells were assembled in an argon-filled glovebox, using either a PP separator or a TiN/BN@PP separator. Electrochemical impedance spectroscopy (EIS) was carried out in the frequency range of 10 mHz to 100 kHz. Linear sweep voltammetry (LSV) evaluations for separator electrochemical stability employed stainless-steel/separator/lithium metal cells, with voltage scans from 3 to 5 V at a rate of 1 mV/s. Constant current charge-discharge tests were conducted at 25°C with a BTS3000 battery testing system (Neware, China) under current densities ranging from 0.2 to 10 C (1C=170 mAh/g) within a voltage window of 2.0-4.2 V (vs. Li⁺/Li). Cyclic voltammetry (CV) measurements were performed using a PARSTAT electrochemical workstation between 2.5 and 4.2 V at scan rates of 0.1-0.5 mV/s. Ionic conductivity (σ) values were derived from EIS data obtained using a stainless-steel/separator/stainless-steel cell, following the Equation (4)[29]:

$$\sigma = \frac{L}{R_b A} \quad (4)$$

Where, R_b is bulk resistance, A is the area of the separator, and L is the separator thickness.

3. Results and discussion

Significant differences in lithium ion deposition behavior between the TiN/BN@PP separator with a blended TiN/BN coating and conventional PP separators are demonstrated in Fig. 1. The uniform coating on the TiN/BN@PP separator regulates lithium-ion distribution during transit and ensures an even arrival at the anode surface. This mechanism promotes the formation of a stable lithium deposition layer at the electrode interface[30]. In contrast, the structural and surface limitations of traditional PP separators cause uneven lithium-ion distribution, leading to localized high concentration regions that promote rapid lithium dendrite growth. Lithium dendrites not only degrade cycling stability but also risk separator penetration, thereby threatening battery safety. Within the composite coating, TiN particles establish a conductive network that reduces interfacial resistance and homogenizes ion transport, while BN particles form a stable protective layer that suppresses rapid lithium dendrite growth owing to their excellent chemical and thermal stability. This composite coating structure significantly improves lithium-ion transport efficiency and enhances overall battery safety and cycle life[31].

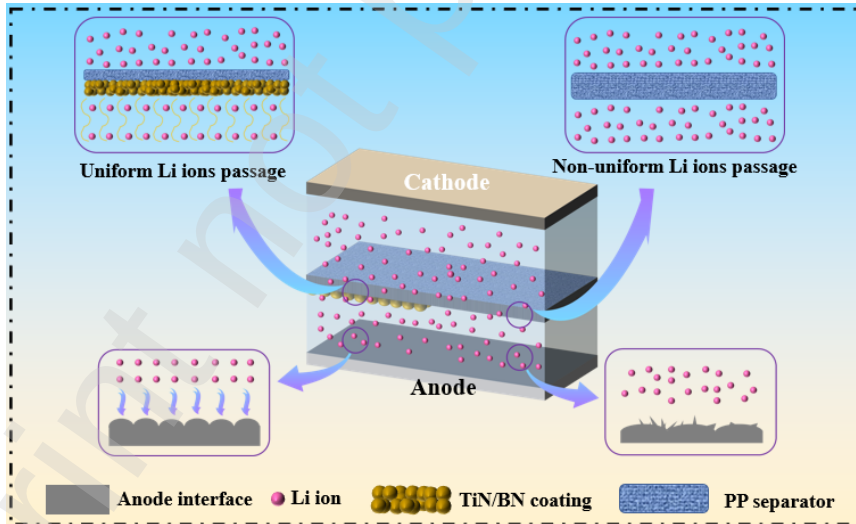


Fig.1 Comparison of lithium ion deposition between TiN/BN@PP separator and PP separator.

We systematically investigated the microstructure and elemental distribution of various separators through SEM and EDS characterization. Fig. 2(a) displays the characteristic irregular porous structure of conventional PP separators. Figs. 2(b-d) show that TiN and BN particles uniformly cover the original porous structure of PP separators, confirming successful formation of dense composite coatings while maintaining structural integrity. The TiN/BN composite coatings optimize the pore structure through

effective void filling and significantly enhance electrolyte adsorption capacity and interfacial wettability of the separator. Subsequently, the cross-sectional morphologies of different separators were analyzed by SEM[32].

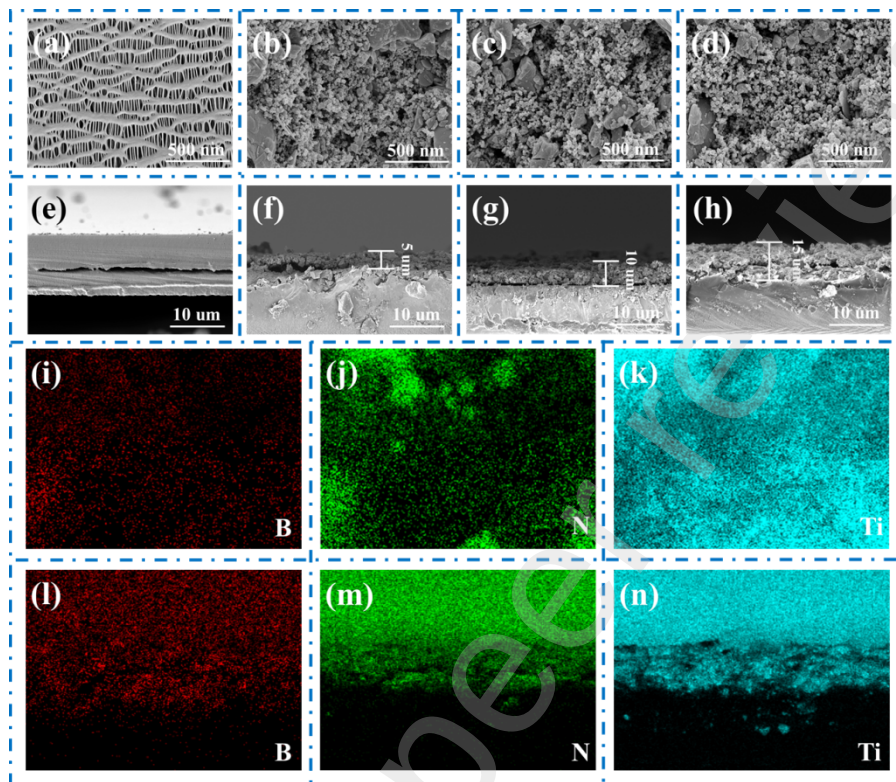


Fig. 2 Surface and cross-sectional SEM images of (a, e) PP, (b, f) TiN/BN@PP-5, (c, g) TiN/BN@PP-10, and (d, h) TiN/BN@PP-15 separators; (i-n) surface and cross-sectional EDS maps of B, N, and Ti elements in the TiN/BN@PP-15 separator.

As shown in Fig. 2(e), the original PP separator retained excellent structural integrity after mechanical shearing. Figs. 2(f-h) reveal tight interfacial bonding between the TiN/BN composite coating and PP basic separator, with no observable delamination or cracks, confirming superior coating adhesion. The composite coatings demonstrate precise thickness control at 5, 10, and 15 μm respectively, verifying both coating uniformity and thickness controllability. These characteristics are critical for enhancing lithium-ion battery cycling stability. Figs. 2(i-k) and Figs. 2(l-n) present elemental distribution patterns on the surface and cross-section of TiN/BN@PP-15. EDS mapping confirms uniform dispersion of boron (B), nitrogen (N), and titanium (Ti) species throughout both surface and cross-sectional areas, without localized concentration gradients. This uniformity confirms complete surface coverage of PP separators by TiN and BN particles. The incorporation of TiN and BN enhances the specific surface area and porosity of PP separators, improving electrolyte retention capacity and promoting enhanced absorption

and ionic conductivity.

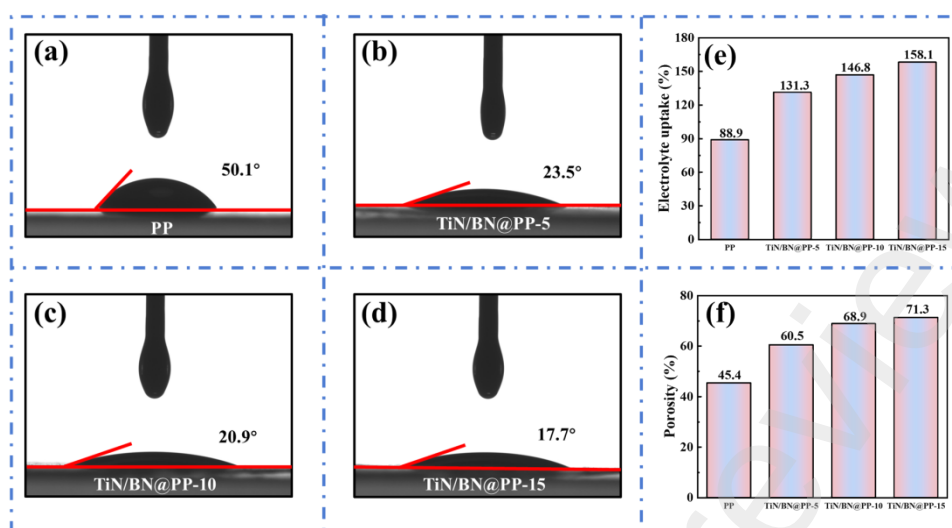


Fig. 3 (a-d) Electrolyte contact angle; (e) electrolyte uptake and (f) porosity of PP and TiN/BN@PP separators.

The wettability of separators directly determines electrolyte absorption efficiency and significantly affects interfacial impedance. To investigate the regulatory effect of TiN/BN composite coatings on separator wettability, four separators were systematically evaluated as shown in Figs. 3a-d. Experimental measurements revealed a contact angle of 50.1° for conventional PP separators. The modified TiN/BN@PP separators exhibited progressively reduced contact angles with increased coating thickness. The TiN/BN@PP-5, TiN/BN@PP-10 and TiN/BN@PP-15 separators showed contact angles of 23.5°, 20.9° and 17.7°, respectively. The TiN/BN@PP-15 separator exhibits a contact angle of less than 20°, representing a 65% reduction compared to the PP separator. This improvement is attributed to the polar surface introduced by TiN and BN particles, which enhances electrolyte wettability. Electrolyte uptake directly reflects absorption capacity. Research indicates that high-porosity separators exhibit superior absorption characteristics, contributing to efficient lithium-ion transport pathways[33]. As shown in Fig. 3e, the PP separator exhibits an electrolyte uptake of 88.9%, while TiN/BN@PP-5, TiN/BN@PP-10 and TiN/BN@PP-15 achieve 131.3%, 146.8% and 158.1%, respectively. The enhanced electrolyte uptake in modified separators facilitates ion transport for stable cycling. Additionally, the TiN/BN@PP separator exhibits no signs of delamination after two hours of immersion in the electrolyte, confirming the excellent chemical stability and strong adhesion of the composite coating. According to Fig. 3f, the porosities of the TiN/BN@PP-5, TiN/BN@PP-10 and TiN/BN@PP-15 separators are 60.5%, 68.9 and 71.3%, respectively, which are significantly higher than the PP separator (45.4%). The thickness of TiN/BN

composite coatings affects separator porosity through specific mechanisms. Increased coating thickness promotes looser stacking of TiN/BN particles, creating additional interconnected voids that enhance effective porosity. These combined effects explain why the TiN/BN@PP-15 separator exhibits superior electrolyte uptake, porosity, and contact angle (indicating improved wettability), and reduced interfacial resistance. The enhanced electrolyte wettability facilitates efficient electrolyte retention, thereby optimizing ion transport kinetics and improving electrochemical performance[34].

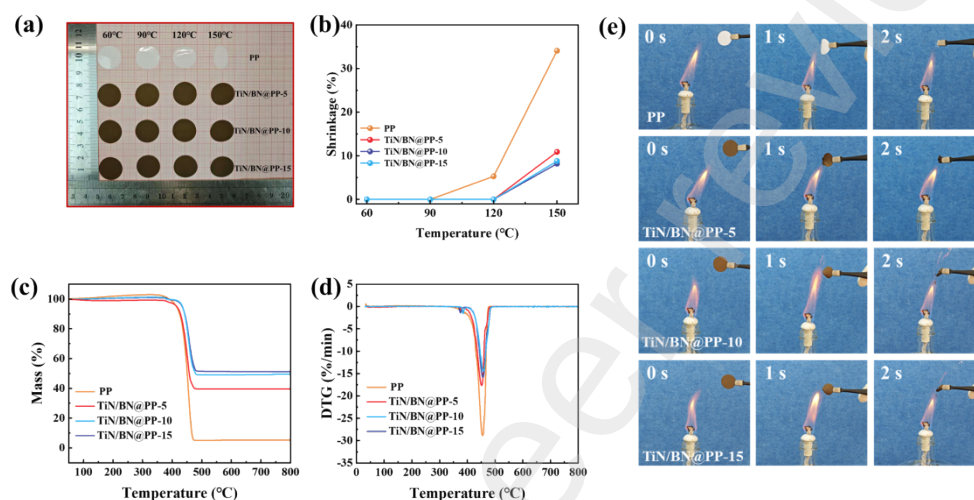


Fig. 4 (a) Images of separators after the heat treatment at different temperatures for 0.5 h; (b) Thermal shrinkage ratio; (c) TG and (d) DTG profiles; (e) Combustion experiment of the separators.

The thermal shrinkage of separators is a critical factor for the safety of batteries, as it reflects the thermal stability. Fig. 4a provides a photo comparison of the different separators responding to heat treatment at temperatures ranging from 60 to 150°C for 30 minutes. The area shrinkage rate is determined by the change in the separator area before and after heating, calculated using the Equation (3)[35]. As shown in Fig. 4a, the PP separator begins to shrink at 120°C and undergoes significant deformation at 150°C, resulting in a thermal shrinkage rate of 34.1%. In contrast, all TiN/BN@PP separators exhibit thermal shrinkage below 1% at 120°C, with no noticeable size reduction. Furthermore, at 150°C, the TiN/BN@PP separators demonstrate markedly better thermal stability than the PP separator under the same conditions. These findings indicate that the TiN/BN@PP separators exhibit superior thermal stability, which is crucial for preventing short circuits caused by thermal shrinkage, thereby enhancing the battery safety. Figs. 4c and d depict the TG-DTG curves of the different separators. The decomposition temperature range for the PP separator is 400-420°C. In contrast, due to the inherent thermal stability and high melting point of TiN and BN, the mass loss of the TiN/BN@PP separators

occurs above 430°C, further confirming the superior thermal stability. Additionally, the comparison of the maximum thermal degradation temperatures between the TiN/BN@PP and PP separators demonstrates that the latter exhibits relatively poor thermal stability[36]. To further evaluate safety performance, combustion tests were conducted (Fig. 4e). The PP separator melted and curled before contacting the flame, igniting immediately upon exposure. In contrast, the TiN/BN@PP separators curled slightly before reaching the flame and self-extinguishing once removed. This behavior demonstrates excellent flame retardancy. These results confirm that the TiN/BN composite coating significantly enhances separator thermal stability by suppressing shrinkage, delaying decomposition, and improving flame retardancy, thereby preventing internal short circuits and improving battery safety[37].

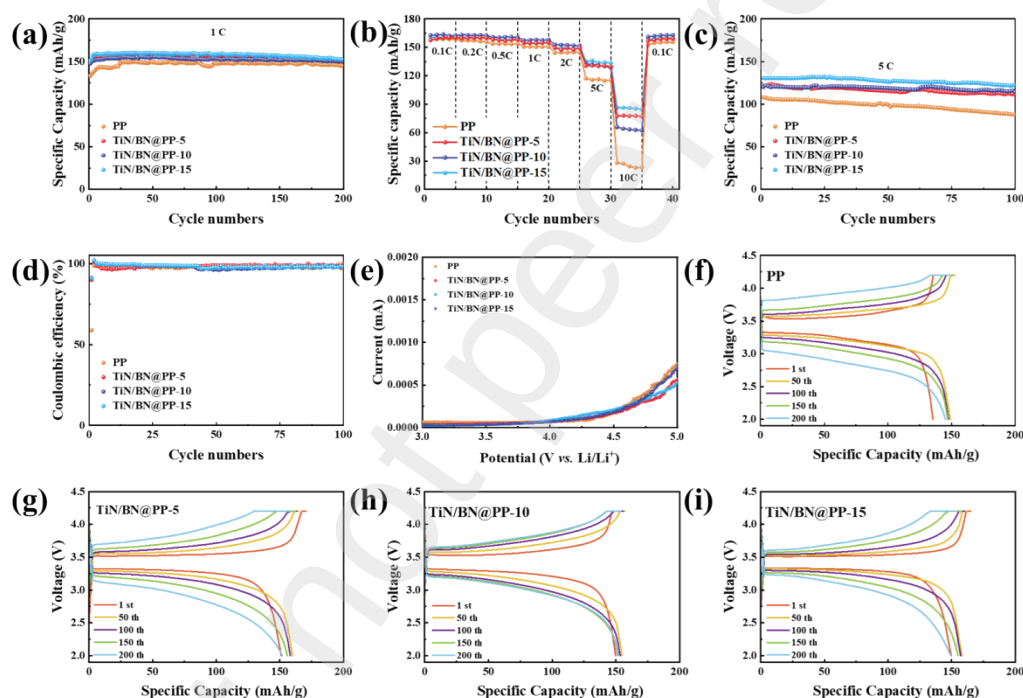


Fig. 5 Electrochemical performance of LiFePO₄/Li batteries with different separators: (a) cycle performance at 1 C; (b) rate performance; (c) 5 C cycling performance; (d) coulombic efficiency at 1 C; (e) LSV curves; (f-i) Charge and discharge curves after different cycles for cells with PP separator, TiN/BN@PP separator-5, TiN/BN@PP separator-10, and TiN/BN@PP separator-15.

Fig. 5a demonstrates the changes in discharge capacity of LiFePO₄/Li cells assembled with different separators over 200 cycles at 1 C. Notably, the cell with the TiN/BN@PP-5 separator showed an initial discharge capacity of 149.7 mAh/g and slightly increased to 149.8 mAh/g after 200 cycles. The cell with the TiN/BN@PP-10 separator exhibited an initial capacity of 147.7 mAh/g that increased to 150.5 mAh/g, while the capacity of the cell with the TiN/BN@PP-15 separator rose from 151.1 to 152.1 mAh/g. In

contrast, the cell with a PP separator started at only 133.3 mAh/g and reached 145.5 mAh/g by the 200th cycle. These data indicate that the discharge capacities of all cells gradually increased with cycling, likely due to the intrinsic activation phenomena of the LiFePO_4 material during cycling. Compared to the traditional PP separator, the incorporation of TiN and BN significantly improved both the discharge capacity and cycle stability[38]. Fig. 5b shows the rate performance of $\text{LiFePO}_4/\text{Li}$ cells assembled with different separators. With increasing discharge rates, all cells experienced a gradual decline in discharge capacity, primarily due to increased electrode polarization under high current conditions. At a current density of 1 C, the cells assembled with the PP, TiN/BN@PP-5, TiN/BN@PP-10 and TiN/BN@PP-15 separators exhibited discharge capacities of 150.6, 154.6, 157.5 and 154.8 mAh/g, respectively. When the discharge rate was increased to 5 C, the capacities dropped to 117.1, 130.8, 132.7 and 135.1 mAh/g, and further decreased to 28.1, 77.5, 65.8 and 86.2 mAh/g at 10 C[39]. The results indicate that, particularly at high discharge rates, the cell with the TiN/BN@PP-15 separator consistently demonstrated superior rate capability. This performance enhancement can be attributed to the effective promotion of rapid lithium-ion transport, reduction in electrode polarization, and optimization of lithium-ion migration pathways by the incorporated TiN and BN particles. Fig. 5c presents the cycling stability at a 5 C discharge rate. The cell employing the PP separator exhibited an initial discharge capacity of 107.9 mAh/g, which declined to 87.9 mAh/g after 100 cycles, reflecting poor high-rate cycling stability. In contrast, the cells assembled with TiN/BN@PP separators exhibited markedly enhanced performance. Specifically, the TiN/BN@PP-15 cell showed an initial discharge capacity of 130.7 mAh/g, which only decreased to 122.3 mAh/g after 100 cycles. Similarly, the cells using the TiN/BN@PP-5 and TiN/BN@PP-10 separators have initial capacities of 121.3 and 120.2 mAh/g, and retained 111.5 and 114.7 mAh/g after cycling[40]. Fig. 5d displays the coulombic efficiencies for the various cells. Although all cells exhibited relatively high coulombic efficiencies, the initial efficiency of the cell with the PP separator was notably lower than those with the TiN/BN@PP separators, implying that the composite coating effectively improves both initial charge transfer kinetics and electrochemical reversibility. Fig. 5e shows the LSV curves for the different separators. The results reveal progressive electrolyte decomposition with increasing voltage, accompanied by a pronounced current response in the high-voltage region. The TiN/BN@PP separator exhibits an electrochemical stability window comparable to the PP separator, but it does not affect the stable operation of TiN/BN@PP separator in the working voltage range. In addition, TiN/BN@PP separator shows better performance in other tests[41]. Figs 5f-i

present the charge-discharge curves of cells assembled with PP, TiN/BN@PP-5, TiN/BN@PP-10, and TiN/BN@PP-15 separators, respectively, at a 1 C discharge rate over multiple cycles. It is evident that the PP separator cell exhibits a marked potential difference between the 1st and 200th cycles. While the thickness of the TiN/BN composite coating increases, this potential difference diminishes, indicating that cell polarization is effectively mitigated. This observation validates the significant role of the TiN/BN@PP separator in optimizing interfacial transport and enhancing stability. These results show that the addition of TiN and BN particles improves the surface structure and wettability of the PP separator, and thus significantly improves the electrochemical performance of the battery[42-44].

Fig. 6a shows the EIS of LiFePO₄/Li cells assembled with different separators. The cell with the PP separator has a charge transfer resistance (R_{ct}) of 151.9 Ω , whereas those with TiN/BN@PP-5, TiN/BN@PP-10 and TiN/BN@PP-15 separators demonstrate progressively reduced R_{ct} values of 127.2, 81.1 and 79.7 Ω , respectively. TiN/BN@PP-15 exhibiting the lowest charge transfer resistance. This suggests that the TiN/BN composite coating effectively enhances the interfacial wettability and conductivity of the separator, thereby reducing the charge transfer resistance and contributing to a more stable electrochemical interface. Furthermore, the lithium-ion diffusion coefficient(D_{Li^+}) was calculated using Equation (5)[45]:

$$D_{Li^+} = \frac{R^2 T^2}{2 \sigma^2 C^2 A^2 n^4 F^4} \quad (5)$$

Where, R is the gas constant, T is the absolute temperature, A represents the electrode area, n is the number of electrons involved in the redox process, F is Faraday's constant, C is the bulk concentration of lithium ions, and σ is the slope of the linear relationship between Z' and $\omega^{-1/2}$.

It is evident that the cell assembled with the TiN/BN@PP-15 separator exhibits the highest lithium-ion diffusion coefficient, which further confirms the advantage of the TiN/BN@PP separator in enhancing the internal lithium-ion transport. As shown in Fig. 6b, the bulk resistance (R_b) of each separator was extracted from EIS, and the ionic conductivity (σ) was calculated using Equation (4). The results reveal that as the thickness of the TiN/BN composite coating increases, the R_b of the separator gradually increases. Specifically, the R_b values for the TiN/BN@PP-5, TiN/BN@PP-10, and TiN/BN@PP-15 separators are 2.63, 2.81 and 3.03 Ω , respectively. Notably, the R_b of all TiN/BN@PP separators are lower than that of the PP separator (4.22 Ω), and the corresponding ionic conductivities

are higher than those of the PP separator. This improvement may be attributed to the role of composite coating, which not only modulate the R_b but also form a microporous architecture within the separator. This structural optimization enhances electrolyte uptake and consequently facilitates lithium-ion transport. Experimental results show that 15 μm is the optimal composite coating thickness since the TiN/BN@PP-15 separator demonstrates the best comprehensive performance[46]. The detailed parameters of the separators are summarized in Table.1.

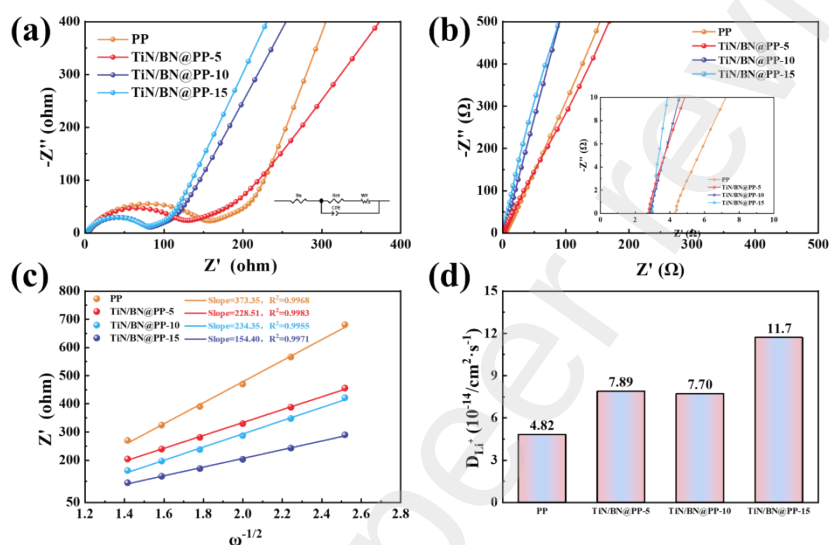


Fig. 6 Electrochemical characterization of cells with different separators: (a) EIS curves; (b) Nyquist plots; (c) the relationship between Z' and the square root of frequency $\omega^{-1/2}$ in the low-frequency region; (d) the diffusion coefficient of lithium ions computed from the derivation of Warburg impedance slope.

Table 1 Bulk resistance; thickness; ionic conductivity; resistance and lithium ion diffusion coefficient of different separators

Sample	R_b (Ω)	D (μm)	σ (mS cm^{-1})	R_{ct} (Ω)	D_{Li^+} ($\text{cm}^2 \text{ s}^{-1}$)
PP	4.22	25	0.209	151.9	4.82×10^{-14}
TiN/BN@PP-5	2.63	30	0.402	127.2	7.89×10^{-14}
TiN/BN@PP-10	2.81	35	0.439	81.1	7.70×10^{-14}
TiN/BN@PP-15	3.03	40	0.466	79.7	11.7×10^{-14}

To investigate the influence of the separator on lithium-ion battery diffusion kinetics, CV tests were performed on $\text{LiFePO}_4/\text{Li}$ cells using different separators, with scan rates systematically varied between 0.1 and 0.5 mV/s. The lithium-ion diffusion coefficient (D_{Li^+}) was determined using Equation (6) based on the Randles-Sevcik Equation[47]:

$$I_p = 2.687 \times 10^5 n^3 D^{1/2} A C \gamma^{1/2} \quad (6)$$

Where, n is the number of electrons transferred in the $\text{Fe}^{2+}/\text{Fe}^{3+}$ redox couple, A is the electrode area, C is the bulk concentration of lithium ions, and γ represents the scan rate.

As depicted in Figs. 7a-d, the peak current (I_p) of all tested samples increased with the scan rate, accompanied by an oxidation peak shift toward higher potentials and a reduction peak shift toward lower potentials. The cell with the TiN/BN@PP-15 separator exhibited the smallest potential difference between the oxidation and reduction peaks, indicating effective suppression of polarization and enhanced reversibility of the electrochemical reaction. Figs. 7e-h display a linear relationship between I_p and the square root of the scan rate ($\gamma^{1/2}$). Calculations using Equation (6) yielded D_{Li^+} values at the reduction and oxidation peaks of 4.11×10^{-11} and $3.12 \times 10^{-11} \text{ cm}^2 \text{ s}^{-1}$ for the TiN/BN@PP-15 cell. 3.05×10^{-11} and $2.10 \times 10^{-11} \text{ cm}^2 \text{ s}^{-1}$ for the TiN/BN@PP-10 cell. 3.3×10^{-11} and $2.24 \times 10^{-11} \text{ cm}^2 \text{ s}^{-1}$ for the TiN/BN@PP-5 cell. These values are all higher than those for the cell assembled with the PP separator (reduction peak: $1.94 \times 10^{-12} \text{ cm}^2 \text{ s}^{-1}$; oxidation peak: $1.51 \times 10^{-12} \text{ cm}^2 \text{ s}^{-1}$). This improvement indicates that the introduction of composite coating enhances lithium-ion diffusion kinetics, thereby accelerating ion transport and improving overall battery performance[48].

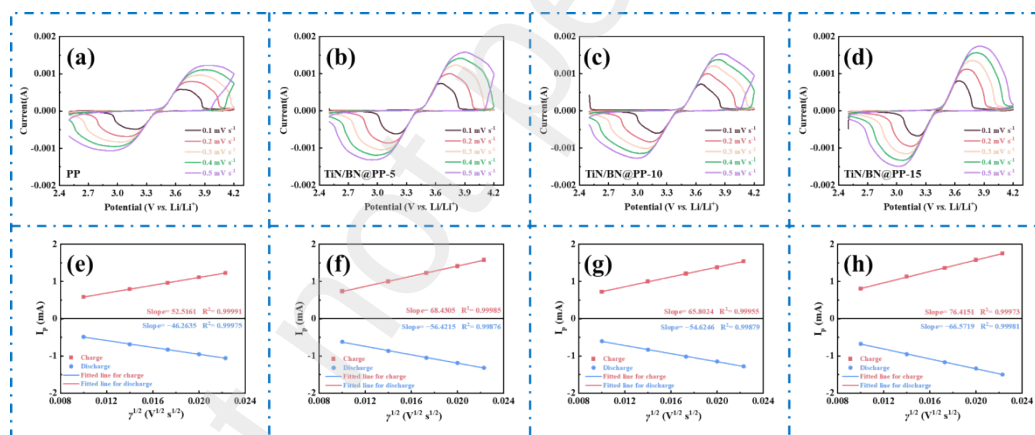


Fig. 7 (a-d) CV curves with different scan rate; (e-h) Fitting curves between the square of peak current and scan rate of PP and TiN/BN@PP separators.

4. Conclusion

In this study, TiN and BN particles were uniformly coated on the PP separator surface to prepare TiN/BN@PP separator. The effects of different thickness coatings on separator characteristics and battery performance were discussed systematically. The results show that TiN/BN@PP-15 separator exhibits excellent wettability, thermal stability and electrochemical properties. Compared with the traditional PP separator, the TiN/BN@PP-15 separator has a high porosity of 71.3%, a high electrolyte

uptake of 158.1% and a low contact angle of 17.7°, which greatly improves the wettability of the separator. In the thermal stability test, TiN/BN@PP-15 separator has a higher initial decomposition temperature, and the shrinkage rate is less than 10% when heated at 150°C for 30 minutes, and part of the structure can be retained after combustion. In the electrochemical test, the first discharge capacity of the battery assembled by TiN/BN@PP-15 separator reached 152.1 mAh/g at 1 C discharge rate, and the capacity retention rate was 100% after 100 cycles. At a high discharge rate of 5 C, the initial discharge capacity achieves 130.7 mAh/g. In addition, the TiN/BN@PP-15 separator assembled battery also has a minimum interface impedance of 79.7 Ω , a high lithium-ion diffusion coefficient of $11.7 \times 10^{-14} \text{ cm}^2 \text{ s}^{-1}$, and a high ion conductivity of 0.466 mS cm^{-1} . This research provides a new idea for the design of high performance and high safety battery separator.

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